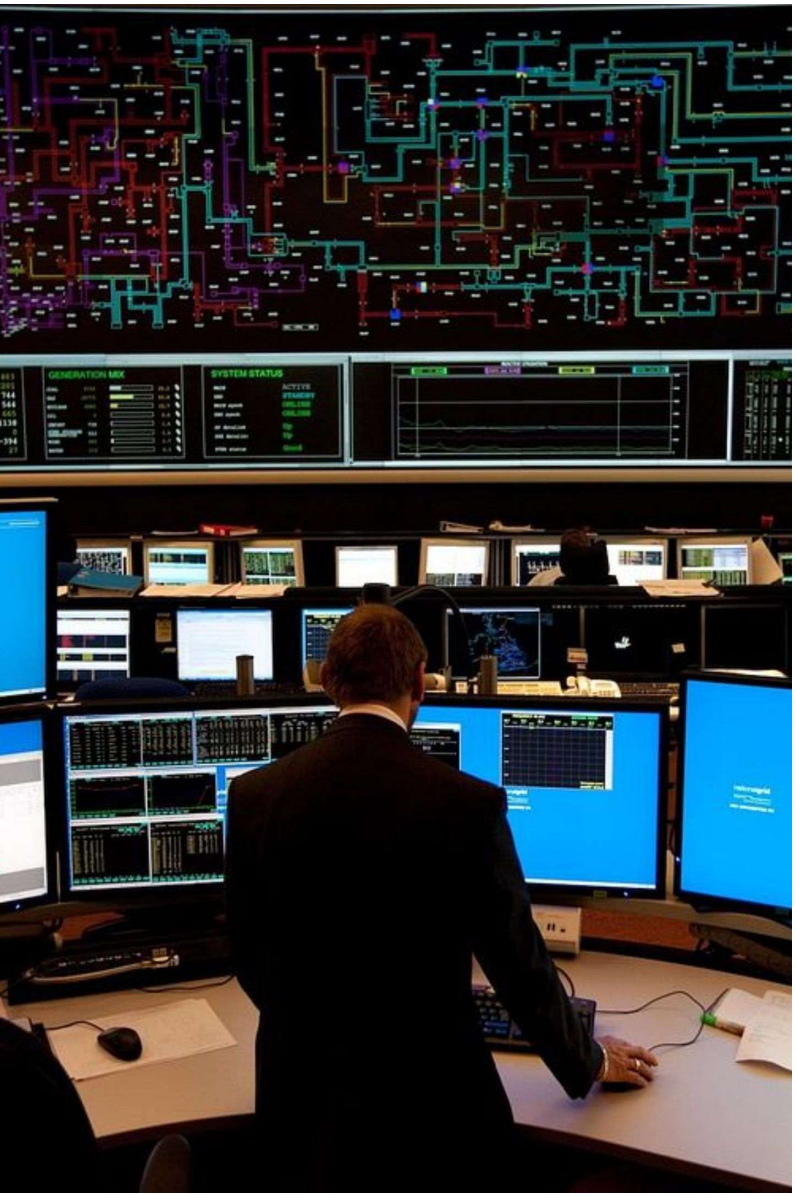
Load Flow Studies: Analysing Power System Stability

A comprehensive introduction to network modelling, bus classification, admittance matrices, and iterative solution methods for modern power systems.



The Core Objective

Generation Balance

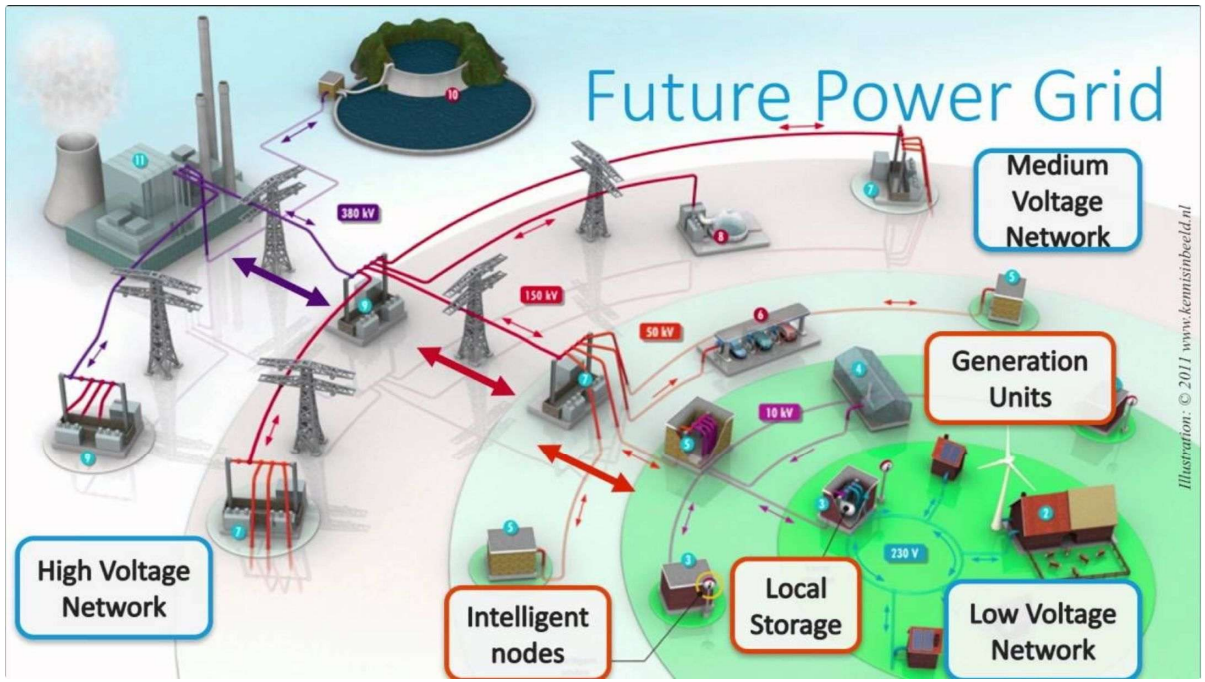
Ensure total generation precisely meets system demand plus all network losses at every operating point.

Voltage Control

Maintain bus voltage magnitudes and angles within prescribed statutory and operational limits.

Thermal Limits

Prevent overloading of transmission lines and transformers to safeguard equipment and system integrity.



Defining the Network

A power system is represented as an interconnected network of **buses** (nodes), **branches** (lines and transformers), and **generation sources**.

Load flow analysis determines the **steady-state voltage magnitudes and phase angles** at every bus — the fundamental unknowns of the system.

These results are essential for both **long-term capacity planning** and **day-to-day operational scheduling**.

POWER SYSTEMS FUNDAMENTALS

What is a bus?

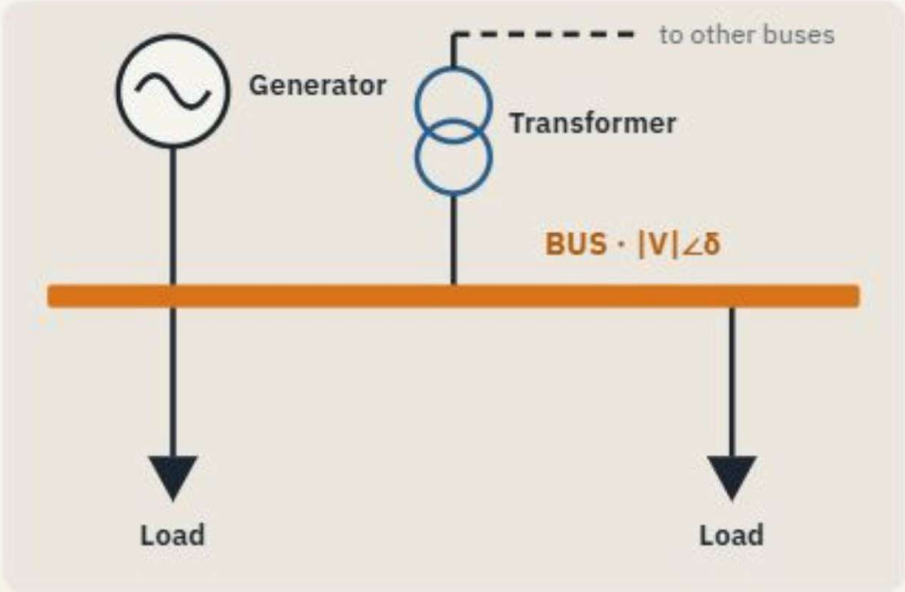
A **bus** is a node where generators, loads, lines and transformers connect. Physically it is a conductor bar in a substation; in analysis it is one point described by four quantities.

|V|
Voltage magnitude
Size of the bus voltage, in kV or per unit

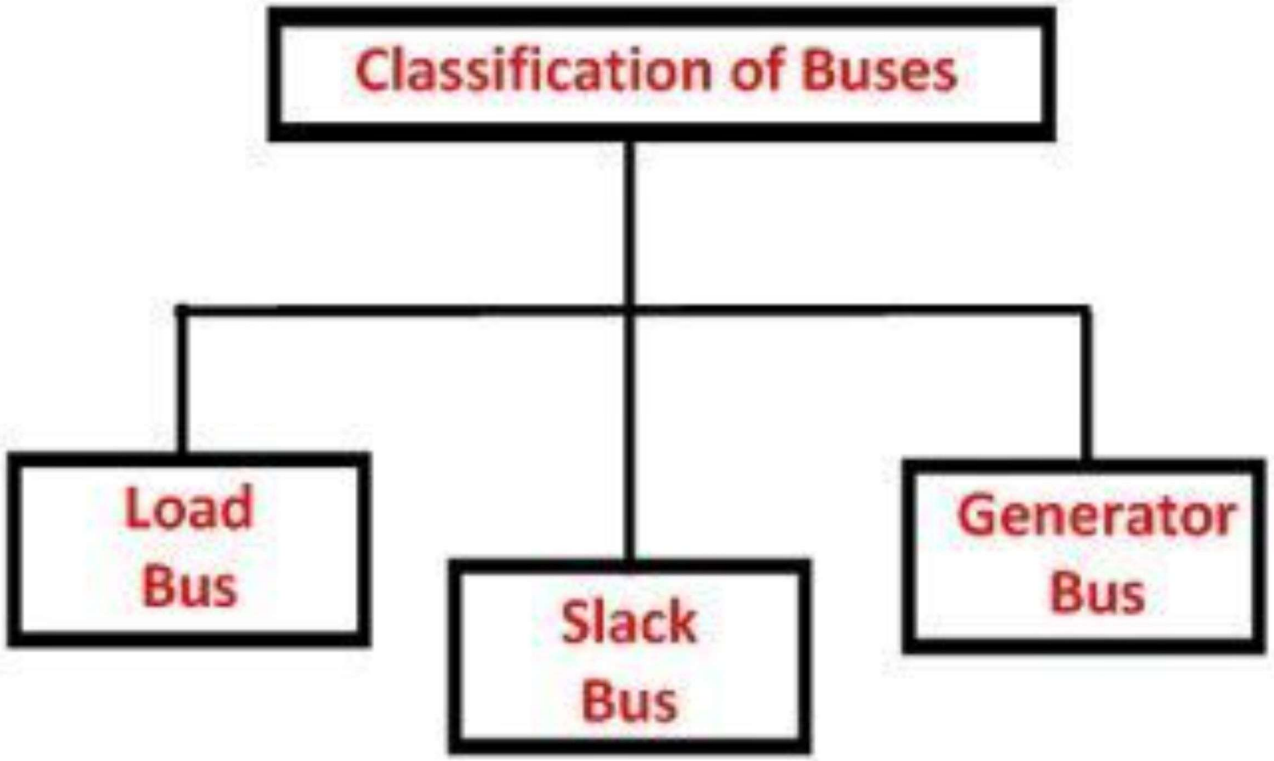
δ
Voltage phase angle
Angle of the voltage relative to a reference bus

P
Real power
Net active power injected at the bus, in MW

Q
Reactive power
Net reactive power injected at the bus, in MVAr



One-line diagram: every element meets at the same node


Classification of Buses
Load Bus
Slack Bus
Generator Bus
Circuit Globe

BUS CLASSIFICATION

The Three Pillars of Bus Classification

E

Slack Bus (|V|δ Bus)

The system reference bus. Both voltage magnitude **|V|** and phase angle **δ** are fixed. Absorbs generation/load imbalances.

Generator Bus (PV Bus)

Generator bus. Real power **P** and voltage magnitude **|V|** are specified. Reactive power **Q** is the unknown.

Load Bus (PQBus)

Load bus. Both real power **P** and reactive power **Q** are specified. Voltage magnitude and angle are the unknowns.

The Slack Bus: The System Balancer

- Only **one** slack bus exists per network — it is the single reference node
- Supplies the deficit between total load plus losses and the sum of all other specified generation
- Voltage magnitude and angle are both **fixed**, making it the computational anchor for all other bus calculations

Why Only One Slack Bus?

Every power system requires exactly one slack bus. Because transmission losses are unknown before solving the load flow, the slack bus absorbs the difference between total scheduled generation and actual system demand plus losses — acting as the real-time balancer.

It also provides the **angular reference** (conventionally set to 0°) against which all other bus angles are measured.

CHAPTER 2

The Admittance Matrix

Building the mathematical backbone of the network model — the nodal admittance matrix, $\mathbf{Y}_{bus}$.

Nodal Admittance Matrix (Y_{bus})

The Y_{bus} matrix is the fundamental mathematical representation of a power network. It relates **bus current injections** to **bus voltages** via the compact expression:

$$I = Y_{bus} \times V$$

Each element encodes the admittance relationships between every pair of buses, capturing both the network **topology** and **impedance characteristics**.

Network Connectivity

Captures which buses are interconnected and the admittance of each branch.

Sparse Structure

Most elements are zero — only connected buses have non-zero off-diagonal entries.

Complex Valued

Elements are complex numbers, representing both conductance and susceptance.

Y*bus* Structure and Properties

Diagonal Elements (Y*ii*)

Self-admittance of bus *i*. Equals the sum of all admittances directly connected to that bus, including shunt elements. Always the largest-magnitude entries.

Off-Diagonal Elements (Y*ij*)

Mutual admittance between buses *i* and *j*. Equal to the *negative* of the admittance of the branch connecting them. Zero if buses are not directly connected.

Symmetry

Y*bus* is **symmetric** (Y*ij* = Y*ji*) for passive networks, halving the storage and computation required.

BUS ADMITTANCE MATRIX

How to build Ybus in five steps

1. **Number the buses**
   Label buses 1 to n; ground is the reference node.

2. **Convert impedances to admittances**
   For every line, $y_{ij} = 1 / z_{ij}$.

3. **Fill the diagonal elements**
   Y_{ii} = sum of all admittances connected to bus i, including shunts.

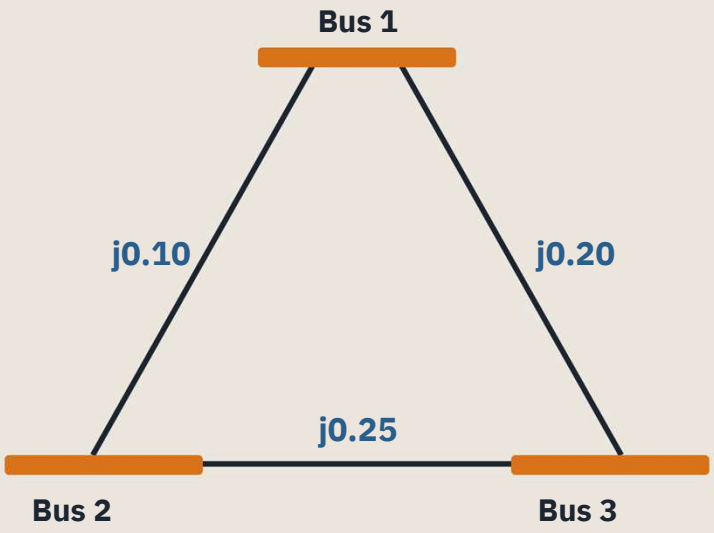
4. **Fill the off-diagonal elements**
   $Y_{ij} = Y_{ji} = -y_{ij}$ between connected buses; 0 if there is no line.

5. **Assemble the n × n matrix**
   Collect the elements into Ybus and check that it is symmetric.

BUS ADMITTANCE MATRIX

Example: a three-bus system

GIVEN



lected.

STEP 1 OF 4

Convert impedances to admittances

$y_{ij} = 1 / z_{ij}$

Line	Impedance z	Working	Admittance y
1 – 2	j0.10	1 / j0.10	$y_{12} = -j10$
1 – 3	j0.20	1 / j0.20	$y_{13} = -j5$
2 – 3	j0.25	1 / j0.25	$y_{23} = -j4$

Remember: 1 / j = −j, so each purely reactive line gives a negative imaginary admittance.

STEP 2 OF 4

Fill the diagonal elements

Y_{ii} = sum of the admittances connected to bus i

Y_{11}

Bus 1 connects to lines 1–2 and 1–3

$y_{12} + y_{13}$
$= -j10 + (-j5)$

$= -j15$

Y_{22}

Bus 2 connects to lines 1–2 and 2–3

$y_{12} + y_{23}$
$= -j10 + (-j4)$

$= -j14$

Y_{33}

Bus 3 connects to lines 1–3 and 2–3

$y_{13} + y_{23}$
$= -j5 + (-j4)$

$= -j9$

STEP 3 OF 4

Fill the off-diagonal elements

$Y_{ij} = Y_{ji} = -y_{ij}$

$Y_{12} = Y_{21}$

Line between bus 1 and bus 2

$-y_{12} = -(-j10)$

= j10

$Y_{13} = Y_{31}$

Line between bus 1 and bus 3

$-y_{13} = -(-j5)$

= j5

$Y_{23} = Y_{32}$

Line between bus 2 and bus 3

$-y_{23} = -(-j4)$

= j4

STEP 4 OF 4

The bus admittance matrix

$$Y_{bus} = \begin{pmatrix} -j15 & j10 & j5 \\ j10 & -j14 & j4 \\ j5 & j4 & -j9 \end{pmatrix}$$

Symmetric
$Y_{ij} = Y_{ji}$ for every pair of buses.

Each row sums to zero
−j15 + j10 + j5 = 0, because there are no shunt elements.

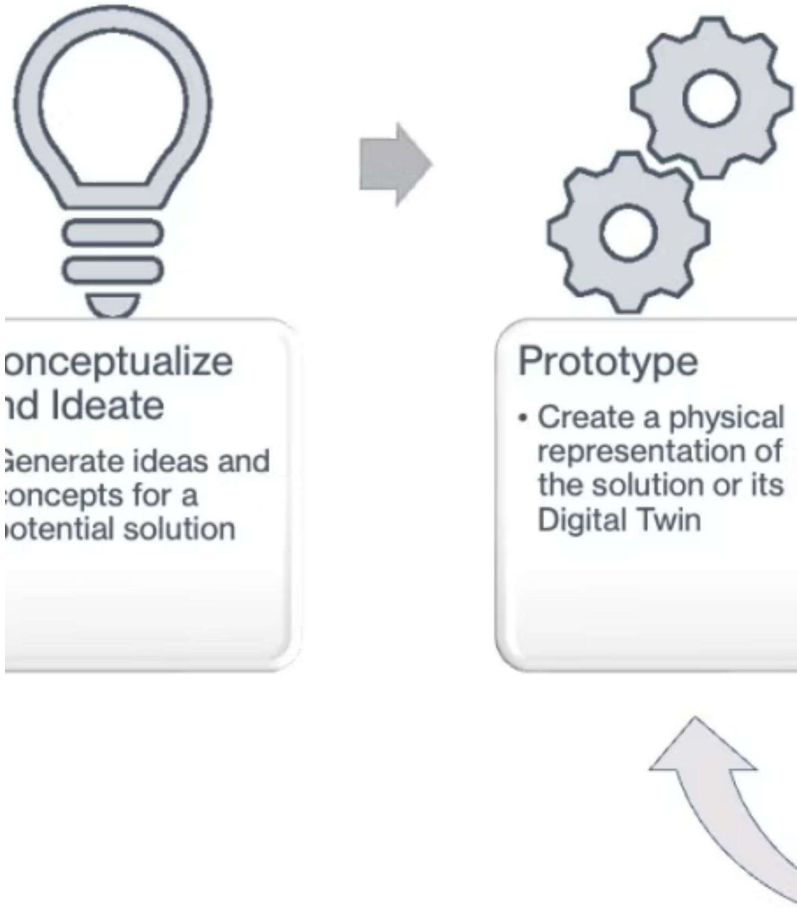
Diagonal dominates
Each diagonal term is the negative sum of the others in its row.

All values in per unit. Diagonal elements highlighted.

CHAPTER 3

Gauss-Seidel Load Flow Analysis

Solving the non-linear power flow equations iteratively — one bus at a time — until the entire network reaches a consistent steady-state solution.



GAUSS–SEIDEL LOAD FLOW

Introduction

Load flow equations are **nonlinear**, so they cannot be solved directly. The Gauss–Seidel method solves them **iteratively**: guess the bus voltages, then correct them one bus at a time until they stop changing.

Starts from a guess

Usually a flat start: every unknown voltage set to $1.0\angle 0°$ pu.

Uses the latest values

Each new bus voltage is used at once when updating the next bus.

Works with Ybus

The bus admittance matrix is the only network data it needs.

Simple but slow

Easy to program and light on memory, but it may need many iterations.

GAUSS–SEIDEL LOAD FLOW

Objective

Given the network (Ybus) and the specified quantities at each bus, find the **voltage magnitude |V|** and **phase angle δ** at every bus.

AT EACH BUS, TWO OF FOUR ARE KNOWN

Bus	Specified	To be found
Slack bus	|V|, δ	P, Q
Load bus	P, Q	|V|, δ
Generator bus	P, |V|	Q, δ

ONCE VOLTAGES ARE KNOWN, COMPUTE

- **Power supplied by the slack bus**
- **Real and reactive power flow in each line**
- **Line losses and total system losses**

GAUSS–SEIDEL LOAD FLOW

Equations

Power injected at bus i: (Voltage at the bus) * (Net Current)

$$P_i - jQ_i = V_i^* \sum Y_{ij} V_j$$

Solving it for V_i gives the update rule:

$$V_i^{(k+1)} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{(V_i^{(k)})^*} - \sum_{j<i} Y_{ij} V_j^{(k+1)} - \sum_{j>i} Y_{ij} V_j^{(k)} \right]$$

Newest values first

Buses already updated (j < i) use their new value $^{(k+1)}$; the rest use the old one.

Generator buses

Find $Q_i = -\mathrm{Im}\{V_i^* \sum Y_{ij} V_j\}$ first, then update V_i and reset $|V_i|$ to its specified value.

Convergence

Stop when $|V_i^{(k+1)} - V_i^{(k)}| \le \varepsilon$ at every bus, e.g. $\varepsilon = 0.0001$ pu.

$$\boxed{\text{Bus Voltage} = \frac{\text{Total current} - \text{Net current outflow}}{\text{Bus admittance}}}$$

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j$$

$$I_i = Y_{ii} V_i + \sum_{j \neq i} Y_{ij} V_j$$

$$Y_{ii} V_i = I_i - \sum_{j \neq i} Y_{ij} V_j$$

$$\boxed{V_i = \frac{1}{Y_{ii}} \left[I_i - \sum_{j \neq i} Y_{ij} V_j \right]}$$

$$I_i = \frac{P_i - jQ_i}{V_i^*}$$

$$\boxed{V_i = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{V_i^*} - \sum_{j \neq i} Y_{ij} V_j \right]}$$

GAUSS–SEIDEL LOAD FLOW

Iteration procedure

1. **Form Ybus and choose a flat start**
   Set every unknown bus voltage to $1.0\angle 0°$ pu; the slack voltage stays fixed.

2. **Update bus voltages in order**
   Apply the Gauss–Seidel equation to buses 2, 3, … n, one after another.

3. **Use each new voltage immediately**
   The updated V_i replaces the old value before the next bus is solved.

4. **Check convergence**
   If any $|\Delta V|$ exceeds ε, start the next iteration from step 2.

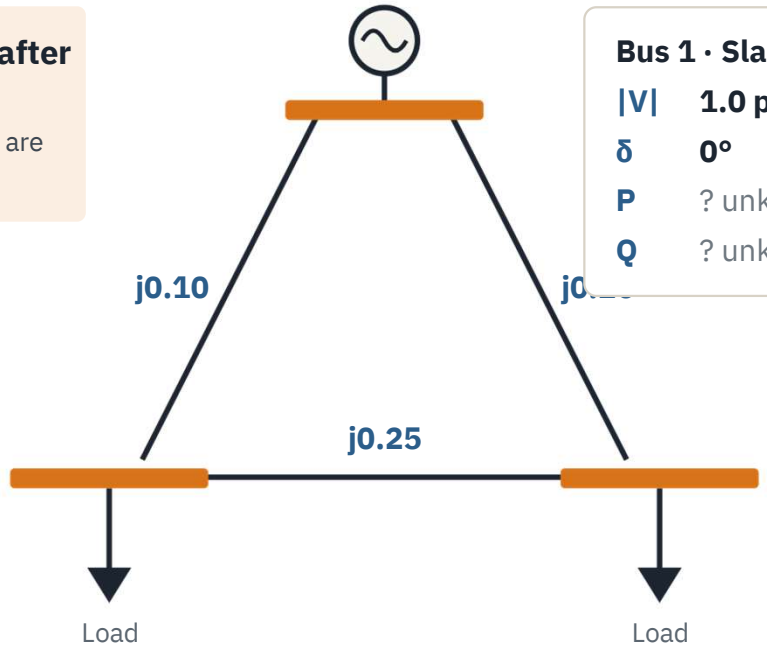
5. **Compute the results**
   Find the slack bus power, line flows and losses from the final voltages.

GAUSS–SEIDEL LOAD FLOW

Example: the three-bus system

Find V_2, V_3 and the slack power after one iteration

Ybus is from the earlier example. P and Q are injected power, negative at loads.



Bus 1 · Slack

|V|	1.0 pu
δ	0°
P	? unknown
Q	? unknown

Bus 2 · Load

|V|	1.0 pu (guess)
δ	0° (guess)
P	−0.5 pu
Q	−0.2 pu

Bus 3 · Load

|V|	1.0 pu (guess)
δ	0° (guess)
P	−0.4 pu
Q	−0.3 pu

Dark: specified **Orange:** initial guess (flat start) Grey: unknown

GAUSS–SEIDEL LOAD FLOW · ITERATION 1

Update bus 2

$$V_2^{(1)} = (1 / Y_{22}) \left[(P_2 - jQ_2) / (V_2^{(0)})^* - Y_{21}V_1 - Y_{23}V_3^{(0)} \right]$$

Term	Value
$(P_2 - jQ_2) / (V_2^{(0)})^*$	$(-0.5 + j0.2) / 1.0 = -0.5 + j0.2$
$- Y_{21}V_1$	$-(j10)(1.0) = -j10$
$- Y_{23}V_3^{(0)}$	$-(j4)(1.0) = -j4$
Sum inside [ ]	$-0.5 - j13.8$
Divide by $Y_{22} = -j14$	$(-0.5 - j13.8) / (-j14)$

RESULT

$V_2^{(1)} = 0.9857 - j0.0357$

$= 0.9864\angle -2.08°$ pu

GAUSS–SEIDEL LOAD FLOW · ITERATION 1

Update bus 3

$$V_3^{(1)} = (1 / Y_{33}) \left[(P_3 - jQ_3) / (V_3^{(0)})^* - Y_{31}V_1 - Y_{32}V_2^{(1)} \right]$$

Term	Value
$(P_3 - jQ_3) / (V_3^{(0)})^*$	$(-0.4 + j0.3) / 1.0 = -0.4 + j0.3$
$- Y_{31}V_1$	$-(j5)(1.0) = -j5$
$- Y_{32}V_2^{(1)}$	$-(j4)(0.9857 - j0.0357) = -0.1429 - j3.9429$
Sum inside [ ]	$-0.5429 - j8.6429$
Divide by $Y_{33} = -j9$	$(-0.5429 - j8.6429) / (-j9)$

Key step: the new $V_2^{(1)}$ is used, not the starting value 1.0.

RESULT

$V_3^{(1)} = 0.9603 - j0.0603$

$= 0.9622\angle -3.59°$ pu

GAUSS–SEIDEL LOAD FLOW · ITERATION 1

Compute the slack bus power

$$P_1 - jQ_1 = V_1^* \left[Y_{11}V_1 + Y_{12}V_2^{(1)} + Y_{13}V_3^{(1)} \right]$$

Term	Value
$Y_{11}V_1$	$(-j15)(1.0) = -j15$
$Y_{12}V_2^{(1)}$	$(j10)(0.9857 - j0.0357) = 0.3571 + j9.8571$
$Y_{13}V_3^{(1)}$	$(j5)(0.9603 - j0.0603) = 0.3016 + j4.8016$
Sum inside [ ]	$0.6587 - j0.3413$
Multiply by $V_1^* = 1.0$	$P_1 - jQ_1 = 0.6587 - j0.3413$

Check: the lines are lossless, so at convergence P_1 must equal the total load of 0.9 pu. The gap means more iterations are needed.

RESULT

$P_1 = 0.6587$ pu

$Q_1 = 0.3413$ pu

Power supplied by the slack bus, using the voltages from iteration 1

GAUSS–SEIDEL LOAD FLOW

After iteration 1

V_1

$1.0\angle 0°$

Slack bus, fixed

$S_1 = 0.6587 + j0.3413$

$V_2^{(1)}$

$0.9864\angle -2.08°$

$0.9857 - j0.0357$

$|\Delta V_2| = 0.0385$

$V_3^{(1)}$

$0.9622\angle -3.59°$

$0.9603 - j0.0603$

$|\Delta V_3| = 0.0722$

Not converged yet

The voltage changes are far above $\varepsilon = 0.0001$ pu, and P_1 is still short of the 0.9 pu total load, so iteration 2 starts from these voltages and repeats the same steps.

Acceleration Factors

In large networks, Gauss-Seidel can converge **very slowly**, requiring hundreds of iterations. An **acceleration factor α** (typically 1.4–1.6) is applied to each voltage update:

$$V_{new} = V_{old} + \alpha \times (V_{calc} - V_{old})$$

This extrapolates the voltage correction, significantly reducing the iteration count. However, values of α above approximately 2.0 can cause **divergence**.

α = 1.0

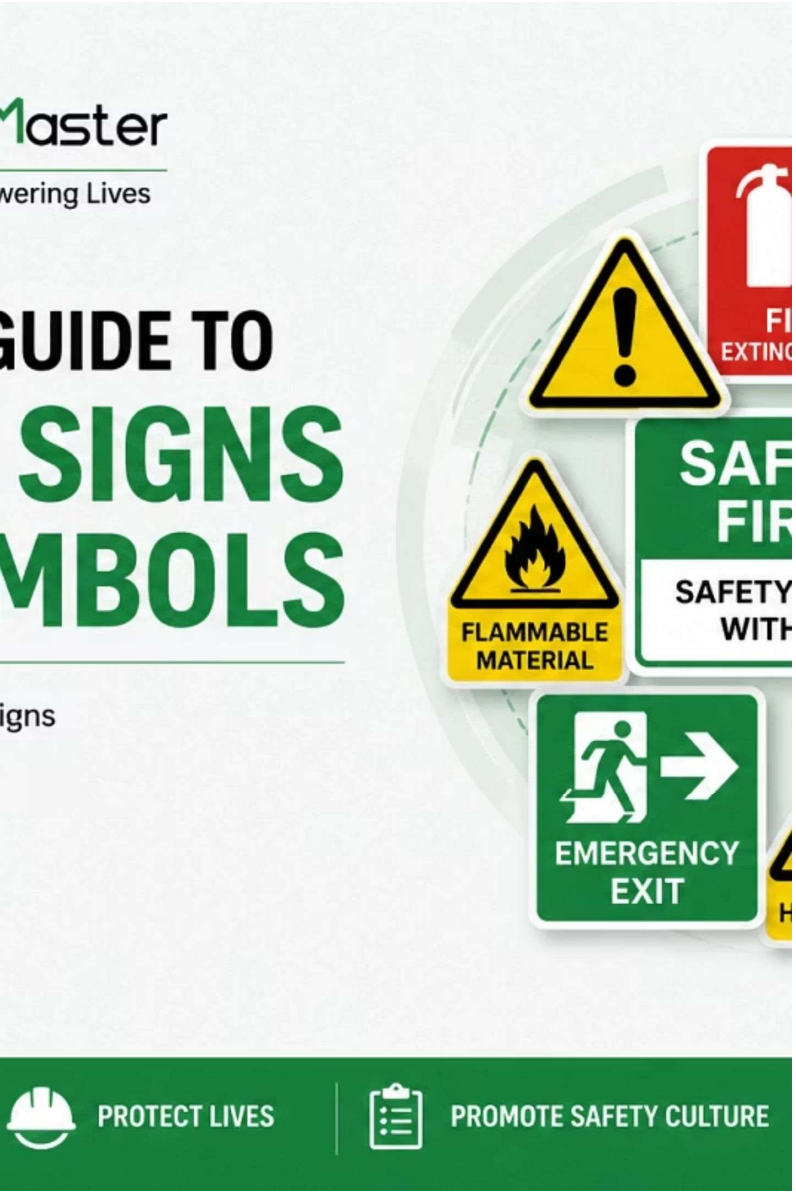
Standard Gauss-Seidel. No acceleration. Slowest convergence.

α = 1.4–1.6

Recommended range. Balanced speed and stability for most networks.

α > 2.0

Risk of divergence. Over-correction destabilises the iterative process.



Limitations and Considerations

Slow Convergence

Significantly more iterations are needed compared to Newton-Raphson, especially for large, heavily loaded networks with many buses.

Negative Reactance Sensitivity

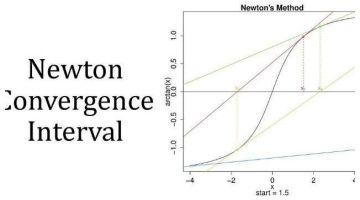
Series capacitors or other elements with negative branch reactances can prevent convergence entirely, requiring special handling.

Solution Uniqueness Risk

For heavily loaded or complex networks, the method may converge to a low-voltage solution or fail to find the operating solution altogether.

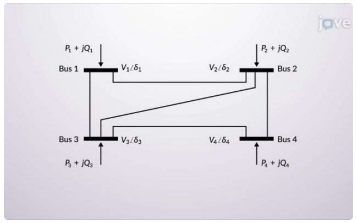
ADVANCED METHODS

Beyond Gauss-Seidel



Newton-Raphson

Uses the Jacobian matrix for quadratic convergence. Superior for ill-conditioned systems and the **industry standard** for large networks.



Fast Decoupled Load Flow

Exploits the near-independence of **P–δ** and **Q–|V|** relationships to decouple the Jacobian, dramatically reducing computation time per iteration.



Method Selection

The right method depends on network scale, conditioning, and speed requirements. Modern energy management systems often employ **hybrid approaches**.

Conclusion: Powering the Future

Load flow analysis is the **cornerstone of reliable grid operation** — from initial network modelling and Y*bus* construction, through iterative Gauss-Seidel convergence, to post-processing of line flows and losses.

Foundation

Y*bus* and bus classification provide the mathematical framework for every network study.

Method

Gauss-Seidel offers a transparent, accessible entry point to iterative load flow solution.

Future

Mastering these techniques is essential for planning the **resilient, low-carbon grids** of tomorrow.

